Imaging Simulation and Learning-Based Image Restoration for Remote Sensing Time Delay and Integration Cameras

Menghao Li, Ziran Zhang, Shiqi Chen, Zhihai Xu, Qi Li, Huajun Feng, and Yueting Chen

***Abstract*— Time delay and integration (TDI) cameras are widely used in remote sensing areas because they capture high-resolution and high signal-to-noise ratio (SNR) images and images in low-light environments. However, the image quality captured by TDI cameras may be affected by many degradation factors, including jitter, charge transfer time mismatch, and drift angle. Moreover, compared with the single-line push-broom cameras and area gaze cameras used in remote sensing, the degraded effect of the TDI camera may accumulate during the charge accumulation process. In this article, we present a fast imaging simulation method for remote sensing TDI cameras based on image resampling that can accurately simulate the degraded image quality affected by different degradation factors. The simulated image pairs can provide a sufficient dataset for modern supervised-learning image restoration methods. In addition, we present a novel network, containing a row-attention block and row-encoder block to help resolve the row-variant blur to resolve the degraded images. We test our image restoration method on the simulated degraded image datasets and real images; the results show that the proposed method can effectively restore degraded images. Our restoration method does not rely on auxiliary information detected by high-frequency sensors or multispectral bands, and it achieves better results than other blind restoration methods.**

***Index Terms*— Deep learning-based image restoration, degraded factors, imaging simulation, time delay and integration (TDI) camera.**

I. Introduction

CURRENTLY, time delay and integration (TDI) cameras are widely used in remote sensing areas. The charge-coupled device (CCD) sensors on the focal plane form a face array, but TDI cameras are a special kind of push-broom camera because they scan the ground object with multiple CCD sensor lines in the face array [1]. The photogenic charge is moved line by line [2], accumulated through the charge transfer process, and then output in the final line to form the final image, which means that ideally, the same ground object is exposed multiple times (the number of times can be called the TDI-CCD stage, which is equal to the CCD rows on the sensor plane). The basic mechanism of the TDI camera is shown in Fig. 1. Although the TDI camera can capture high-resolution and high signal-to-noise ratio (SNR) images, it is more sensitive to degraded factors because of the TDI effect, which means that the degraded factors may accumulate during the photogenic charge integration process. The TDI camera degradation mechanism differs from other remote sensing imaging methods, including single-line push-broom imaging and plane-array gaze imaging [3]. Therefore, constructing a TDI camera imaging simulation model and restoring degraded images captured by TDI cameras is of great importance. In this article, we focus on three types of degradation factors, e.g., jitter, charge transfer time mismatch, and drift angle, which are closely related to the imaging mechanism of the TDI camera.

1) The attitude jitter is a notorious degradation factor caused by aircraft engine vibration, the fluctuation of the environmental temperature, and the operation control, as shown in Fig. 2(a). It is hard to eliminate and ultimately leads to the blur and distortion of the captured images [4], [5], [6].
2) As the TDI camera exposes the same object several times through photogenic charge accumulation, the photogenic charge transfer time between adjacent CCD rows must match the ground motion velocity [1], [7], [8]. To be more specific, the photogenic charge transfer time should obey

$$t = \frac{d}{f}/(v/H) \tag{1}$$

where t is the ideal photogenic charge transfer time from one stage to the next, d is the pixel size, f is the focal length of the optical system, v is the velocity of the aircraft carrying the TDI camera, and H is the height between the aircraft and the object. The terrain height and velocity fluctuation both cause photogenic charge transfer time mismatch, thus degrading the images [shown in Fig. 2(c)].
3) The TDI camera requires the image movement direction to be consistent with the integration direction. The difference angle between these two directions is called the drift angle [8], [9], which may be caused by the Earth's rotation or aircraft maneuvering. The drift angle may result in a modulation transfer function (MTF)

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The authors are with the State Key Laboratory of Extreme Photonics and Instrumentation and the College of Optical Science and Engineering, Zhejiang University, Hangzhou 310027, China (e-mail: chenyt@zju.edu.cn).

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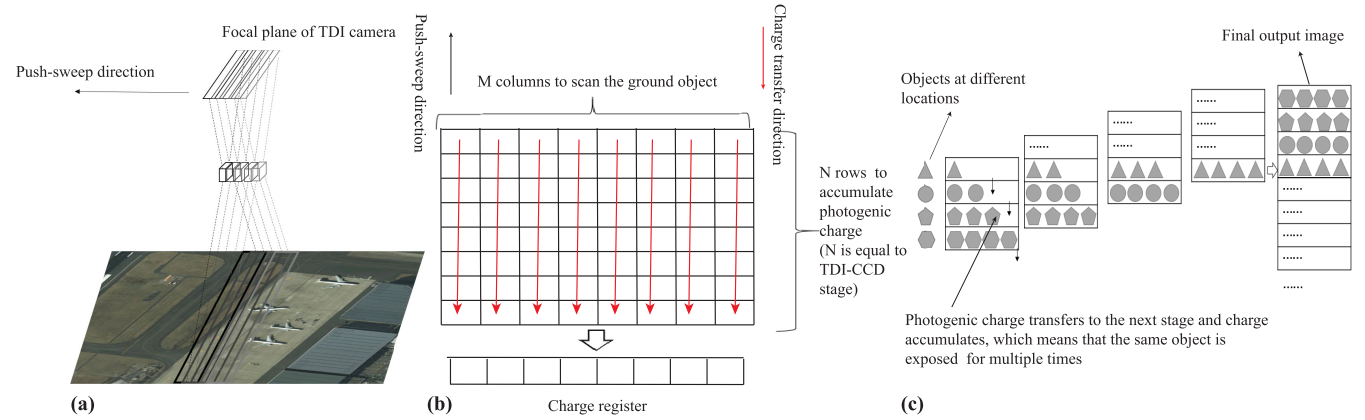


Fig. 1. Imaging mechanism of a TDI camera. (a) TDI cameras are a special kind of push-broom camera. When capturing images, the camera moves along with the aircraft. (b) Schematic of the TDI camera. (c) Photogenic charge accumulation process of the TDI camera (the TDI-CCD stage is set to four). The photogenic charge should transfer to the next stage in time to ensure that the same object is exposed multiple times.

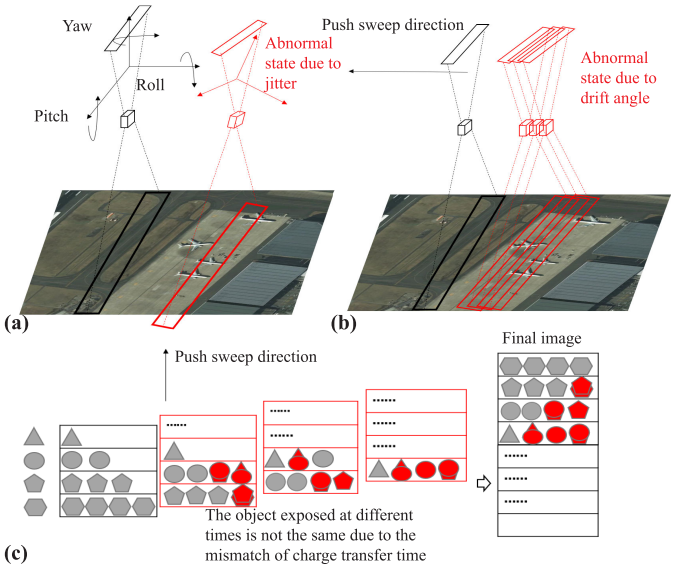


Fig. 2. Common degradation factors of the TDI camera. (a) Aircraft carrying the TDI camera may vibrate along three axes, i.e., roll, pitch, and yaw axis. (b) Photogenic charge integration direction does not match the push-sweep direction, i.e., drift angle degradation factor. (c) If the charge transfer time does not match the aircraft flight velocity, the photogenic charge for integration does not represent the same object, and the final image will blur.

drop and degrade the image quality [8], [10] [shown in Fig. 2(b)].

Previous remote sensing simulation methods have mainly focused on single-line push-broom cameras or array gaze cameras, which neglect the charge accumulation process or push-broom imaging mode of TDI cameras [11], [12], [13], [14], [15]. Traditional imaging simulation methods for TDI cameras commonly use a single point spread function (PSF; some researchers prefer to call it blur kernel instead of PSF) or an analytic MTF to process each row [3], [16], [17]. However, when coupled degradation factors occur, the PSF or MTF is complicated to obtain or even unanalytic. Moreover, previous work considered only one of these degradation factors instead of all coupled degradation factors together. Thus, it is vital to propose a new simulation method for TDI cameras.

Image restoration has been a fundamental and rapidly developing problem in image processing [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32]. It has gained increasing attention in remote sensing areas, whose main tasks include deblurring, denoising, and defogging [11], [33], [34], [35], [36], [37], [38], [39]. For this article, we focus on deblurring and correcting the geometric distortion caused by the degradation factors of the TDI camera.

Some remote sensing image restoration methods consider the following mathematical model [11], [33]:

$$g = h \otimes o + n \tag{2}$$

where g is the observed degraded image, o is the latent clean image, n represents the noise, h is the PSF, and $\otimes$ represents the convolutional operator. However, it is not suited for TDI degradation factors. According to previous research [3], [16], a degraded image captured by a TDI camera has different PSFs for each row, which means that the PSFs are spatially variant; therefore, the mathematical model for the degraded image captured by the TDI camera is given as follows:

$$g(\boldsymbol{x}) = h(\boldsymbol{x}) \otimes o(\boldsymbol{x}) + n \tag{3}$$

where $\boldsymbol{x}$ represents the pixel coordinate in image space. The degraded image suffers from not only blur but also distortion [40], [41], making the image restoration task harder than traditional deblurring tasks. Some previous work to restore remote sensing images uses model-based methods, which show impressive results but usually take a long time to iterate [3], [35], [42]. Most model-based methods commonly assume spatially invariant blur, which may not be appropriate for coupled degraded factors. Moreover, previous work to restore degraded images captured by TDI cameras depends on assist information acquired by high-frequency sensors or parallax images [42], [43], [44], [45], [46], [47], which cannot be acquired for every aircraft carrying TDI cameras.

The main contribution of this manuscript is given as follows.

1) We propose a TDI camera imaging simulation model that can accurately simulate different degradation factors in a single process compared with previous work. Moreover, our method can simulate complicated coupled

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degradation factors for which we cannot obtain an analytic degradation expression because it is based on the Monte Carlo method. It is an image-to-image simulation method beneficial for producing paired-image datasets for learning-based methods. Furthermore, based on the imaging simulation model, we analyze degraded TDI camera image quality in detail.

2) In some situations, we can obtain only the degraded images without other information to assist image restorations. We propose a blind learning-based method that can effectively deblur and correct the geometric distortion of degraded images without assistance information. Moreover, considering the degraded TDI image features, we propose a row-encoder and row-attention block, which can efficiently improve the image restoration quality. The experiment shows that the proposed method outperforms model-based restoration methods and other learning-based methods. To our knowledge, we are the first to use a learning-based method to effectively restore degraded TDI images.

The remainder of this article is organized as follows. In Section II, we briefly introduce previous TDI camera imaging simulation and restoration methods for remote sensing images. In Section III, we introduce our TDI imaging simulation method in detail, and in Section IV, we introduce our image restoration method. Section V shows the experimental results, and this article is concluded in Section VI.

II. Related Works

A. TDI Camera Imaging Simulation Models

Traditional remote sensing degradation simulation models use a PSF or an analytical modulation function to degrade clean images [11], [12], [13], [14]. In particular, Wahballah et al. [8] proposed several analytical MTFs based on the camera's pointing accuracy and angular velocity. However, these simulation methods ignore the fact that the degraded images captured by TDI cameras are commonly spatially variant blurs, so they may not be suited for imaging simulations.

A comprehensive TDI camera imaging simulation should consider the push-broom mode and charge accumulation mechanism. The traditional method used PSF convolution to degrade the clean image row by row [3], [17]. The PSF of each row is first calculated according to the relative movement, and then, the blurry rows of different moments are summed to simulate the charge integration process. This method can simulate the jitter effects, but it is difficult to compute the analytic PSF when the degradation becomes complicated. Zhou et al. [16] proposed a vibration MTF model based on statistical data and then processed clean images row by row. However, this model considers the precise assembly imaging structure of each component, which is difficult to obtain in common cases. Pan et al. [47] used a continuous dynamic shooting mode to simulate the push-broom mode and refine the detected jitter curves. However, this model first uses multispectral band images to estimate the jitter curves. Image interpolation methods are also used to simulate the jitter effects [15], [48]. However, they consider only the across-track jitter and ignore the charge accumulation process. Other simulation methods consider the imaging link, including the camera's rotation, light sources, scene reflection, and electronic response [49], [50]. They are comprehensive simulation methods but sophisticated and not suitable for generating image pairs.

B. Image Restoration Models for Remote Sensing Images

Degraded images captured by TDI cameras suffer from both blur and geometric distortion. Previous work to restore degraded images first estimates the degradation factor parameters. For example, jitter curves are estimated using high-frequency measuring equipment, subpixel parallax image matching, or deep learning-based methods [15], [42], [43], [44], [45], [46], [47], [48]. Once the jitter curves are determined, they use image resampling to compensate for the jitter curves and correct the geometric distortion. However, image resampling models cannot deblur images. For deblurring tasks, traditional methods first estimate the PSF. Cao et al. [11] proposed a neural network to refine the initially unidirectional PSF, and Zhang et al. [35] estimated the PSF using a local binary pattern prior. However, degraded images captured by TDI cameras are spatially variant blurred, and a single PSF is not adequate for expressing degradation. Wu et al. [3] and Liu et al. [42] considered the spatially variant blur and estimated PSFs in different image regions based on the detected jitter curves. Then, they deblurred the degraded images row by row using the Richard–Lucy method. However, deconvolution methods do not perform well on geometric correction tasks. To summarize, few previous works have considered deblurring the image and correcting the geometric distortion. Most of them rely on assistive information to carry out image restoration.

Recently, with the rapid development of learning-based image restoration methods in computer vision areas, learning-based methods have also been widely adopted to restore degraded remote sensing images. Zhu et al. [34] and Burdziakowski [51] used a GAN-based neural network, and Wulich and Kopeika [52] used a multilevel feature fusion network to deblur motion-blurred remote sensing images. Shu et al. [36] proposed a network to restore turbulence-degraded images. However, these methods do not specifically consider the imaging mechanism of TDI cameras. In addition, although researchers constantly propose impressive learning-based image restoration methods, they mainly focus on degraded factors such as camera shaking encountered in daily life instead of specific factors of TDI cameras [53], [54]. Whether they can be applied to TDI-degraded image restoration remains to be explored.

III. TDI Camera Imaging Simulation Model

A. Moving Window Image Resampling Model

The spectral response of the TDI camera depends on the band number on the chip. We start by discussing a single spectral TDI camera simulation, which can be easily extended to multispectral TDI cameras. The label images used in the following discussion are all converted to gray to represent the

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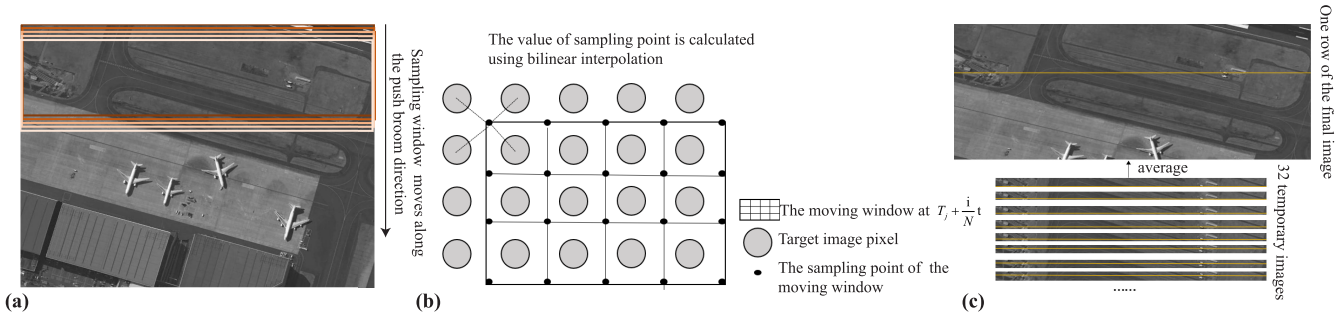


Fig. 3. Our TDI camera imaging simulation method. (a) Sampling window with $w' \times h'$ (w' is the column number of CCD and h' equals the TDI-CCD stage) scans the label image from top to bottom. (b) For each time (i/N) t in one photogenic charge transfer time, the sampling window samples the label image using bilinear interpolation. The sampling values of all moments are averaged to form the temporary image. (c) Final image row is the average of the temporary images' corresponding row. For example, assume that the TDI-CCD stage is 32, the ith row of the final image is the average of the 32nd row of the i_{th} temporary image, the 31st row of the $(i+1)$th temporary image, and the 30th row of the $(i+2)$th temporary image until the 1st row of the $(i+31)$th temporary image.

original response within one band. In the remainder of this article, we assume that the image row is in the across-track direction and the image column is in the along-track direction.

The mathematical model of TDI camera imaging can be expressed as follows:

$$g(m,n,k\times\triangle T) = g(m-1,n,(k-1)\times\triangle T) + \int_{(k-1)\times\triangle T}^{k\times\triangle T}\iint_{0}^{X,Y} e(x,y,t)dxdydt \tag{4}$$

where g represents the photogenic charge at one pixel location, m and n represents the CCD row and column on the sensor plane, respectively, $\triangle T$ represents the exposed time (photogenic charge transfer time), k represents the kth charge transfer time, X and Y represent the size of one pixel, and $e(x,y,t)$ represents the received energy of one point. The equation indicates that the photogenic charge of one pixel comes from two parts: the charge transferred from the previous CCD line and the photogenic charge generated in the current charge transfer time.

We can use the Monte Carlo method to approximate the integration

$$g(m,n,k\times\triangle T) = g(m-1,n,(k-1)\times\triangle T) + \frac{1}{N}\sum_{i}^{N}(e(t_i,x_i,y_i)\times X\times Y\times\triangle T) \tag{5}$$

where t_i, x_i, and y_i are the random sampling points from $(k-1)\times\triangle T$ to $k\times\triangle T$. N is the number of sampling points. Then, one pixel of the final output image $f(p,q)$ can be formulated as

$$f(p,q) = \sum_{k=1}^{\text{TS}}\frac{1}{N_k}\sum_{i}^{N_k}(e(t_i,x_i,y_i)\times X\times Y\times\triangle T) \tag{6}$$

where p and q represents the pixel coordinate, TS represents the TDI-CCD stage, and N_k represents the number of points sampled from $(k-1)\times\triangle T$ to $k\times\triangle T$. Our simulation is based on this equation, which can be considered a variant of the Monte Carlo method. The sampling rate should be selected according to degraded parameters and Shannon's sampling theorem to avoid the aliasing effect. In most cases, we find that a sampling rate of 40 is adequate for simulation accuracy.

We use a label image to represent the received energy and use an image resampling method to simulate TDI camera imaging [see Fig. 3(a)]. Then, we can use the pixel value interpolated by the label image to represent the sampled value $(e(t_i,x_i,y_i)\times X\times Y\times\triangle T)$ in (5). Assume that the size of the label image is $w\times h$; then, a sampling window with size $w'\times h'$, where w' represents the number of sensor columns on the focal plane and h' represents the number of sensor rows on the focal plane (i.e., TDI-CCD stage), scans the label image from top to bottom. In one photogenic charge transfer period, a time series is formed based on stratified sampling

$$\left[T_j+\frac{0}{N}\Delta t, T_j+\frac{1}{N}\Delta t, T_j+\frac{2}{N}\Delta t,\ldots,T_j+\frac{N-1}{N}\Delta t\right]$$

where T_j represents the jth photogenic charge transfer time, $\triangle t$ represents the photogenic charge transfer time, and N is the sampling rate.

For each moment in the sampled time series, the sampling window samples the image using bilinear interpolation. This process is shown in Fig. 3(b). Then, all sampling values in the photogenic charge transfer period are averaged to form a temporary image. The temporary image represents the photogenic charge accumulated in one period, and the second part is on the right of (5). Finally, all these temporary images are averaged according to their photogenic charge transfer time and form the final output image, shown in Fig. 3(c). Notably, the temporary images are averaged instead of summed because we should match the label image average pixel values to the simulated image average pixel values, which are also used in other TDI camera simulation methods [17], [48]. It can also be assumed that the sampling value energy is (1/TS) of the label image, where TS represents the TDI-CCD stage.

As the aircraft carrying the TDI camera is moving when capturing images, intrinsic velocity should be included in the simulation. Under ideal imaging conditions without other degradation factors, the sampling window should accurately move down one row within one charge transfer time. For the sampled time $(i/N)\Delta t$ within one photogenic charge

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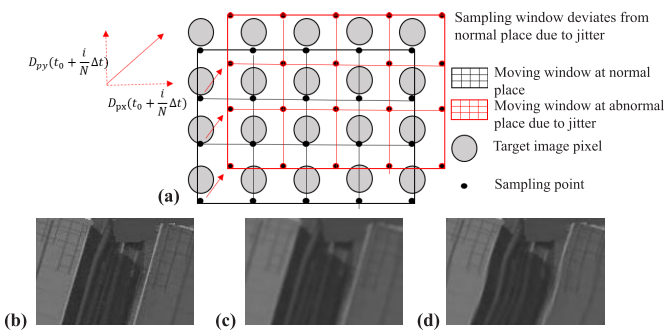


Fig. 4. Jitter curve simulation mechanism. (a) Sampling window deviates from the normal position at the sampled moment. (b) Label image. (c) Degraded image affected by low-frequency jitter. (d) Degraded image affected by high-frequency jitter. We take part of the original image to make the distortion and blur more visually obvious.

transfer time

$$\left[T_j + \frac{0}{N}\Delta t, T_j + \frac{1}{N}\Delta t, \ldots, T_j + \frac{N-1}{N}\Delta t\right]$$

the sampling window moves down for (i/N) pixels. Every row of the TDI camera will obtain the same PSF because of the intrinsic velocity, as mentioned in previous research [3], [17].

B. Simulate Different Degraded Factors

1) Simulating the Jitter Curves: The jitter curve is a degraded factor that receives the most attention because it is hard to eliminate. As mentioned in previous research [15], [43], [52], we can represent the jitter curves using the sum of several sinusoidal functions with different parameters, i.e.,

$$D_p(t) = \sum_{i=1}^{N} A_{pi} \sin\left(2\pi f_{pi} t + \phi_{pi}\right) \tag{7}$$

where $D_p(t)$ represents the total amplitude of the jitter curves in the p-direction at time t, A_{pi} represents the amplitude, f_{pi} represents the frequency, and ϕ_{pi} represents the initial phase.

Jitter occurs in three directions: pitch, roll, and yaw. In image coordinates, the jitter finally degrades the image in the row and column directions. We simulate the jitter by adding additional movements to the current place when resampling images. For one sampled time $t_0 + (i/N)\Delta t$ of the sampling time series, the sampling window deviates from the origin position by $((\boldsymbol{D_{px}(T_j + (i/N)\Delta t)}, \boldsymbol{D_{py}(T_j + (i/N)\Delta t)})$, as shown in Fig. 4.

2) Simulation of the Photogenic Charge Transfer Time Mismatch: As mentioned above, the photogenic charge transfer time should match the ground velocity to capture the same object. If the photogenic charge transfer time is slower (faster) than the standard time, the photogenic charge collected by the sensor does not focus on the same object, and the image will become blurred and shrink (stretch). We simulate this effect by adjusting the intrinsic velocity. In the ideal case, the sampling window moves down by 1 pixel in one photogenic charge transfer time. In photogenic charge mismatch situations, the intrinsic velocity will become $(1 + \alpha)$ pixels in one photogenic charge transfer time, where α is defined as the mismatch factor; a negative α represents a slower transfer time and a positive α represents a faster transfer time. The simulated images degraded by the mismatch shown in Fig. 5, which have similar results as in [7] and [8].

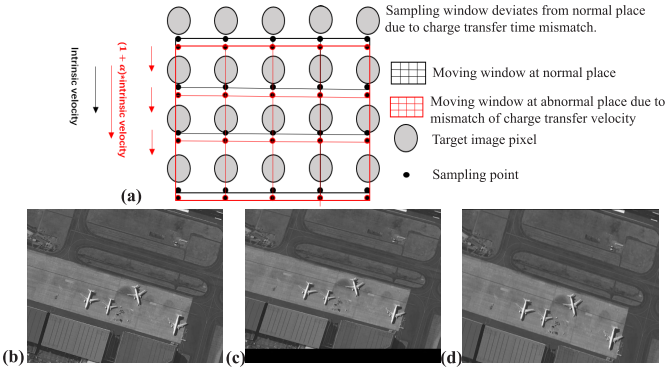


Fig. 5. Photogenic charge transfer time mismatch simulation mechanism. (a) Sampling window's intrinsic velocity changed to $(1 + \alpha)$, and the same object was not exposed for the same time. (b) Label image. (c) Degradation image where the intrinsic velocity is 10% faster than normal (the black part at the bottom indicates that the sampling window moves beyond the label image). (d) Degradation image where the intrinsic velocity is 10% slower than normal.

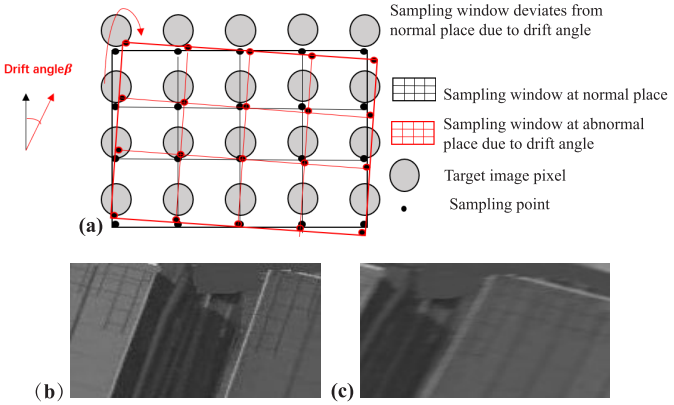


Fig. 6. Drift angle simulation mechanism. β is set to 10°. (a) Sampling window rotates by β due to the drift angle. (b) Label image. (c) Degraded image affected by drift angle.

3) Drift Angle Simulation: As the drift angle varies relatively slowly in one image capturing time, we assume that the drift angle remains the same in one simulation process. We simulate the drift angle by rotating the sampling window by a β angle, which is shown in Fig. 6. The degraded image is not only distorted but also obviously blurred [8].

4) Simulating Coupled Degraded Factors: In real situations, the degradation factors mentioned above may occur together, and it is relatively difficult to obtain an analytic expression for the coupled degradation factors. In addition, as the factors are coupled, simulating each factor separately in sequence is not accurate. However, our simulation method can simulate all the coupled factors in one simulation process. We should adjust the sampling window in each sampling point according to only the degradation parameters, as shown in Fig. 7(a). Specifically, we first rotate the sampling window according to the drift angle, set the intrinsic velocity according to α, and then rotate the sampling window according to

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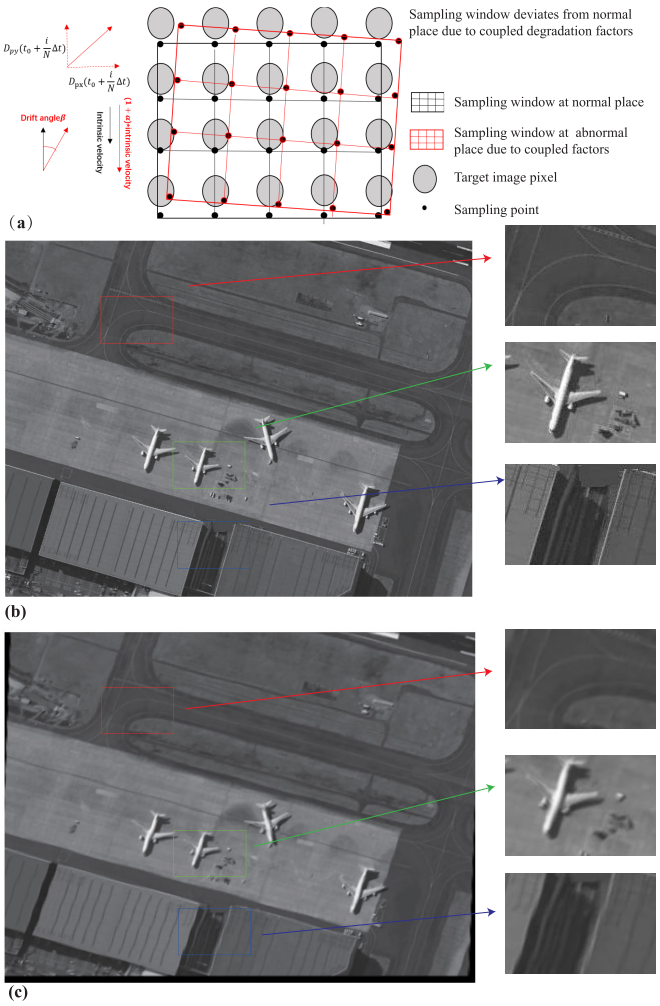


Fig. 7. Coupled degradation factors simulation mechanism. (a) Sampling window adjusts according to the degradation factors' parameters. (b) Label image. (c) Simulated image affected by coupled degradation factors. The jitter curves are $D_{px} = 2\sin(2\pi \times 300 \times t) + \sin(2\pi \times 400 \times t + 1) + \sin(2\pi \times 1000 \times t + 2)) + \sin(2\pi \times 1200 \times t + 3)$ and $D_{py} = 2\sin(2\pi \times 300 \times t + 4) + 1.2\sin(2\pi \times 415 \times t + 5) + \sin(2\pi \times 1000 \times t + 6) + 0.8\sin(2\pi \times 1200 \times t + 7)$, the intrinsic velocity is 1% faster than normal, and the drift angle is 1°. The charge transfer time is 61.996 μs, and the TDI-CCD stage is 32.

$(\boldsymbol{D_{px}(T_j + (i/N)\Delta t), D_{py}(T_j + (i/N)\Delta t)})$. The simulation result is shown in Fig. 7(b) and (c). We present our simulation algorithm as Algorithm 1.

Also, it is relatively easy to extend the simulation for multispectral TDI cameras. The multispectral response TDI camera has the following structure, as shown in Fig. 8. We use Algorithm 1 for different bands in sequence and fuse different degraded bands to form the final multispectral images. The pixel interval of different bands should be considered. We conclude our simulation algorithm in Algorithm 2. The simulation results are included in the Supplementary Materials.

IV. Restoration Method

The degraded images captured by the TDI camera have the following features.

Algorithm 1 Simulation of Degraded Images Captured by the TDI Camera

Require: Label image Q (Image size is $w \times h$),
ideal photogenic charge transfer time $\triangle t$,
TDI-CCD stage(TS), jitter curves $D_{px}t$, $D_{py}(t)$
photogenic charge mismatch factor α,
drift angle β,
sampling rate N.

Ensure: Degraded image O with the same size as the label image

1: initialize:
Rotate around the center of the sampling window by β
Set $velocity_{intrinsic}$ $1+\alpha$ pixel per the photogenic charge transfer time.
Set t_{start}. (This parameter is useful for multispectral spectral simulation. The default value is set to 0.)

2: begin:
For each photogenic charge transfer time T_j ($T_{j+1} - T_j = \triangle t$) until the top of the sampling window meets the bottom of the label image:
Form a time sequence using stratified sampling $[\boldsymbol{T_j + \frac{0}{N}\Delta t, T_j + \frac{1}{N}\Delta t, \ldots\ldots T_j + \frac{N-1}{N}\Delta t}]$
Add t_{start} elementwise to the time sequence.

3: $i = 0$

4: **while** $i < N$ **do**

5: The sampling window moves by $(D_{px}(T_j + \frac{i}{N}\triangle t), D_{py}(T_j + \frac{i}{N}\triangle t) + velocity_{intrinsic} \times \frac{i}{N})$ in the image coordinate, and resamples the image using bilinear interpolation, the sampling parts outside the label image are set to zero.

6: Average the sampling value to form a temporary image with size $w \times TS$

7: Add the temporary image to the final image according to the sampling window position.

8: **end while**

9: Output the final simulated degraded image

1) Different degradation factors are coupled together, making the PSF hard to estimate, and traditional deconvolution methods may not be suitable.
2) Each row of the final degraded image is the summation of degraded slices with the number of TDI-CCD stages, which complicates the degradation.
3) The jitter with a frequency lower than the fundamental aircraft frequency, the photogenic charge transfer time mismatch, and the drift angle all produce geometric distortion. Notably, the restoration methods should not only consider deblurring but also correct distortion.

Due to degradation complexity, traditional methods, including image resampling and estimating the PSF of the whole image, are not suitable. In most practical applications, we can obtain only a single blurred image and know only the approximate range of degraded parameters. Therefore, we adopt the end-to-end blind restoration method. As deep neural networks have a powerful fitting ability, we adopt deep learning methods to restore complicated degraded TDI camera images.

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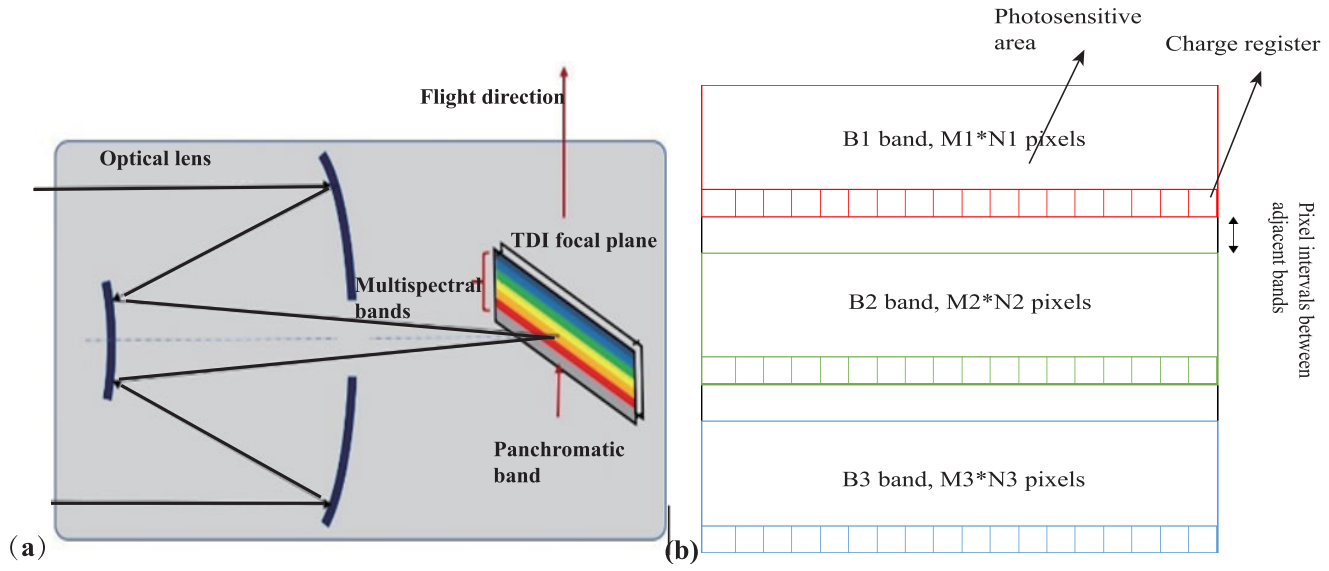


Fig. 8. (a) Structure of multispectral response TDI cameras. (b) Schematic drawing of multispectral bands.

Algorithm 2 Simulation of Multispectral Degraded Images Captured by the TDI Camera

Require: Different bands of label image $Q_1, Q_2 \ldots\ldots Q_N$. Pixel interval between adjacent bands $P_0, P_1 \ldots\ldots P_{N-1}$. Other parameters in Algorithm 1.

Ensure: Degraded image O with the same size as the label image

1: initialize:
$pixel_{interval} = 0$; $t_{start} = 0$
2: begin:
For i_{th} band of the label image Q_i:
3: Set t_{start} to pixel $pixel_{interval} \times \triangle t$.
4: Begin Algotihm1 for this band with t_{start}.
5: $pixel_{interval} += P_{i-1}$
6: Fuse different bands to form the final multispectral images.

Our restoration method assumes that the specific jitter curves, the photogenic charge transfer time mismatch, and the drift angle are not previously known. However, we know the approximate range of the degradation parameters, which is a common situation when the aircraft does not carry high-frequency equipment. The approximate range is used in the row-encoder block discussed later.

We adopt a variant of the UNet architecture [55]; the UNet architecture is a simple but efficient method for image restoration. Our network is shown in Fig. 9. The encoder and decoder are the superposition of basic blocks. To be more specific, the number of basic blocks in the network's encoders is [1, 1, 1, 28] (from larger feature maps to small feature maps), and the number of basic blocks in decoders is [1, 1, 1, 1] based on previous research [25], [31]. We introduce other special parts of the network in detail.

A. Row-Encoder and Row-Attention Blocks

As the influence of degraded factors will accumulate, we can obtain some heuristic structures for our network. The influence of jitter curves is averaged in the final image due to the sum of the temporary images. We can use the mean absolute amplitude (MAA) of the jitter curves to represent the degree of blur for each row of the final image, i.e., the degree of blur of the final image's jth row can be calculated using

$$\mathrm{MAA}(j) = \frac{1}{\mathrm{TS} \times N} \sum_{j=1}^{\mathrm{TS}} \sum_{i=1}^{N} \left| \mathrm{jitter_curves}\left(T_j + \frac{i}{N} \triangle t \right) \right| \tag{8}$$

where N is the sampling rate within one photogenic charge transfer time and TS is the TDI-CCD stage.

Moreover, considering the deviated pixel caused by the photogenic charge transfer time mismatch, if the mismatch is not corrected in time, the deviated pixel will grow linearly with image row growth. All these factors are shown in Fig. 10, which shows an obvious row-variant blur. We are inspired by this and propose two blocks: the row-encoder block and the row-attention block.

1) *Row-Encoder Block:* We concatenate seven row varying weight maps with the degradation image input, which leads the neural network to be aware of the low-frequency jitter curves. To be more specific, six varying weight maps are generated using sinusoidal functions whose frequency is randomly generated in the approximate frequency range and the phase is set to $\mathbf{[0, 0, (\pi/3), (\pi/3), (2\pi/3), (2\pi/3)]}$. When testing, the sinusoidal function frequency is set to the mean of the approximate frequency range, and the phase is handled as above. Another weight map whose weight grows gradually with the image row is also concatenated. The row encoder is shown in Fig. 11(a).
2) *Row-Attention Block:* As the image degradation varies by row, inspired by the spatial attention block, we propose a row-attention block to lead the network to allocate different amounts of attention to different rows, i.e., a more blurred row receives more attention than a less

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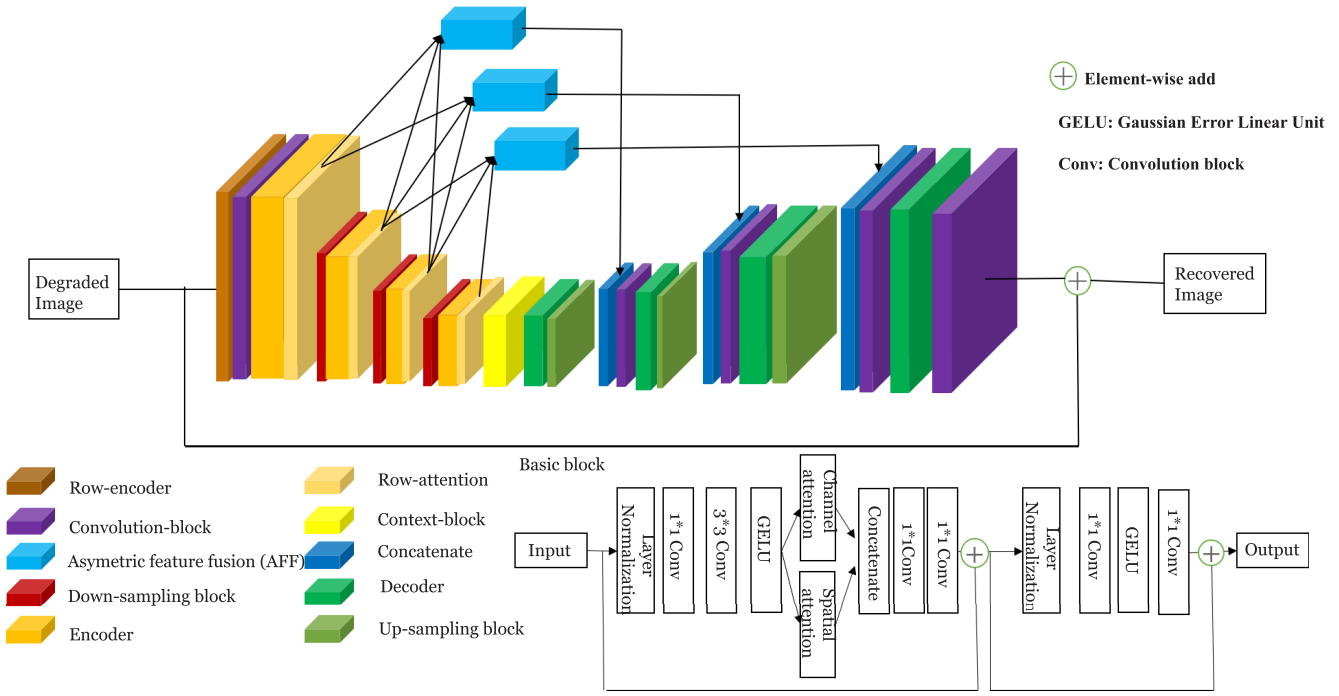


Fig. 9. Proposed network architecture. The details of some blocks are shown in Figs. 11 and 12.

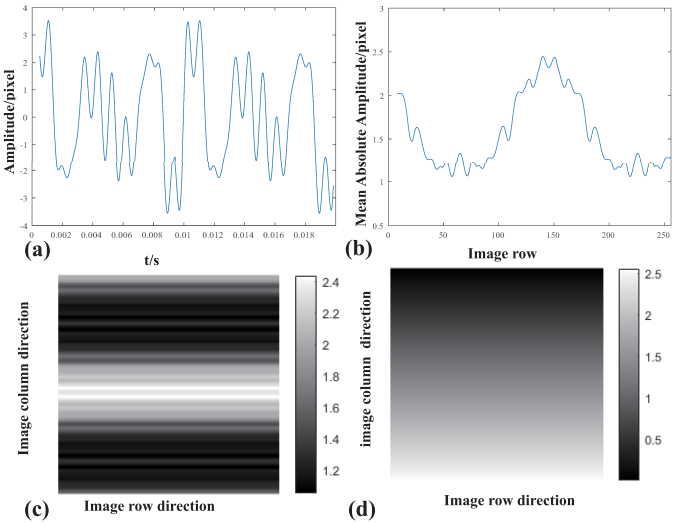


Fig. 10. (a) Simulated jitter curves $2\sin(2\pi \times 300 \times t) + 1.2\sin(2\pi \times 400 \times t + 1) + 1\sin(2\pi \times 1000 \times t + 2) + 0.8\sin(2\pi \times 1200 \times t + 3)$, the sampling time rate is 1.54 μs. (b) Mean absolute jitter amplitude of final images each line (256), the exposed time is set to 61.996 μs and the TDI-Sequence is set to 32. Different rows have different jitter amplitudes, indicating that the degree of blur varies within each row. (c) Absolute mean amplitude of jitter curves in the image. (d) Deviated pixel caused by the photogenic charge transfer time mismatch; the mismatch ratio is set to 1%.

blurred row. The attention is adjusted during network training; this block is shown in Fig. 11(b).

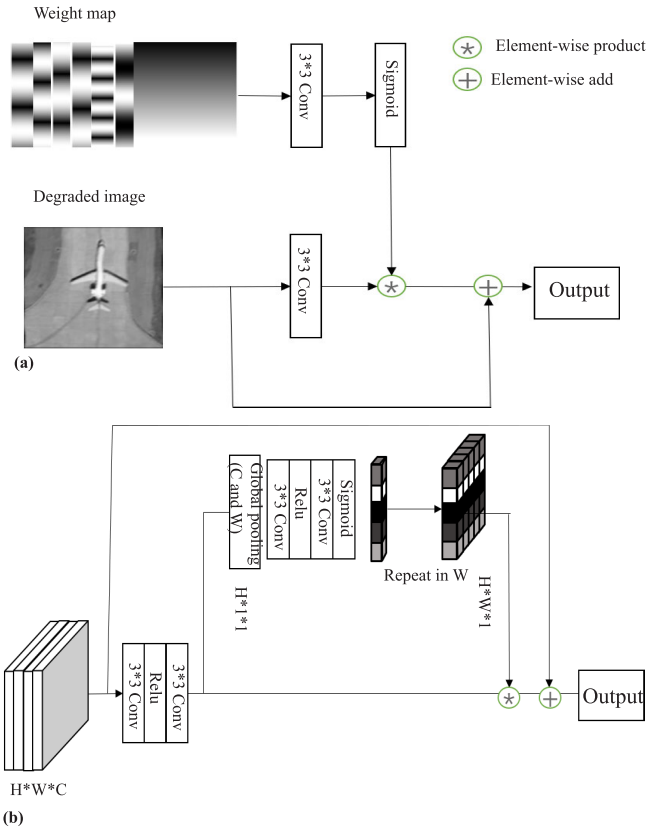


Fig. 11. Details of (a) row-encoder block and (b) row-attention block.

B. Channel Attention Block and Spatial Attention Block

The attention block can refine the extracted features, focusing the network more on the important parts of the feature map. It is found that the attention block impressively improves the image restoration task [56], [57]. The channel attention block has the following expression:

$$F_{\text{out}} = \sigma\left(f_{\text{Ex}}\left(f_{\text{Sq}}(F_{\text{in}})\right)\right) \odot F_{\text{in}} \tag{9}$$

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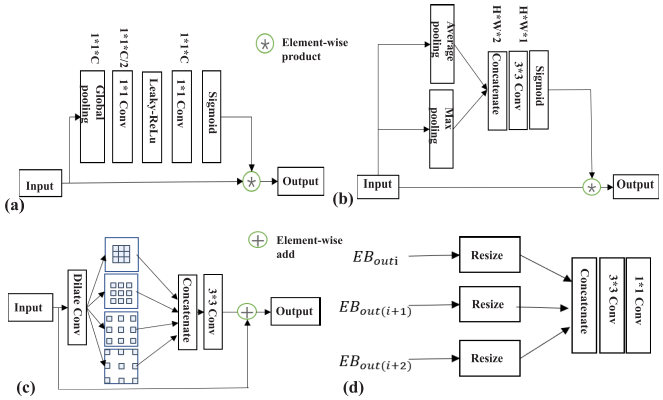


Fig. 12. Details of (a) channel attention block, (b) spatial attention block, (c) context block, and (d) AFF block.

where F_{in} represents the input feature, F_{out} represents the output feature, $\odot$ represents the elementwise product, f_{Ex} represents the excitation process, f_{Sq} represents the squeeze process, and σ represents the sigmoid function. In our implementation, we use global pooling and dimension-reduction convolution as the squeeze process and dimension-increase convolution as the excitation process, as shown in Fig. 12(a).

The spatial attention block has the following expression:

$$F_{\text{out}} = \sigma\left(f^{3\times3}\left(\left[\text{Avg}(F_{\text{in}}); \text{Max}(F_{\text{in}})\right]\right)\right) \odot F_{\text{in}} \tag{10}$$

where Avg represents the average operation across the channel dimension, Max represents the max-pooling operation across the channel dimension, and $f^{3\times3}$ represents the convolution module with a kernel size of 3×3, as shown in Fig. 12(b).

C. Context Block

The context block comprises several dilated convolution blocks and a final fusion block, and we put it into the minimum scale between the encoder and the decoder, similar to a previous network [58]. It has the following mathematical expression:

$$F_{\text{out}} = F_{\text{in}} + f^{3\times3} \times \left(\left[f_{\text{dil}=1}(F_{\text{in}}); f_{\text{dil}=2}(F_{\text{in}}); f_{\text{dil}=4}(F_{\text{in}}); f_{\text{dil}=8}(F_{\text{in}})\right]\right) \tag{11}$$

where F_{in} represents the input feature and F_{out} represents the output feature. $f^{3\times3}$ represents a convolution operation with a filter size of 3×3, and $f_{\text{dil}=a}$ represents a dilated convolution operation with a dilation factor equal to a (in our implementation, the dilated convolution operation has a 3×3 kernel size). The context block can help increase the network receptive field and thus help handle blur caused by a large PSF. This block is shown in Fig. 12(c).

D. Asymmetric Feature Fusion Block

Traditional UNet fuses only features from coarse-scale subnetworks, and we adopt the asymmetric feature fusion (AFF) block to allow information flow from different scales [25]. Thus, each network decoder can obtain different-scale features, which may improve the restoration performance. AFF has the following expression:

$$F_{\text{out}} = f^{1\times1}\left(f^{3\times3}\left(\left[\text{EB}_{\text{outi}}; EB^{*}_{\text{out}(i+1)}; \text{EB}^{*}_{\text{out}(i+2)}\right]\right)\right) \tag{12}$$

where EB_{outi} represents the ith encoder output and EB* indicates that this encoder output should resize to the same size as EB_{outi} for concatenation. This block is shown in Fig. 12(d).

E. Loss

Our network loss function consists of two parts, the Charbonnier loss [59] and the edge loss, i.e.,

$$\begin{aligned} L &= L_{\text{char}}(y_*, y) + \lambda L_{\text{edge}}(y_*, y) \\ L_{\text{char}} &= \sqrt{||y_* - y||^2 + \sigma^2} \\ L_{\text{edge}} &= \sqrt{||\Delta y_* - \Delta y||^2 + \sigma^2} \end{aligned} \tag{13}$$

where y_* represents the label image and y represents the restored image. Δ represents the Laplacian operator and $\sigma = 0.001$ is a constant in all the experiments. L_{char} helps to recover the image blurring facts, L_{edge} helps to correct the distortion, and λ is set to 0.03 according to the experiment. The choice of lambda is discussed in Section V.

V. Experiment and Discussion

A. Analysis of Simulated Degraded Image Quality

To further verify the effectiveness of our proposed imaging simulation method, we implement it using the C++ language in Visual Studio 2019. At a certain sampling point, we can split the sampling window into different patches and resample the label image in parallel on different CPUs. We use 12 11th generation Intel Core[1] I7-11700 processors in parallel. For an image with a size of 256×256, a sampling rate of 40, and TDI-CCD stage 32, the simulation method takes approximately 7 s.

We set the charge transfer time to 61.998 μs according to [43], which is the parameter of the Yaogan-26 satellite [accurate charge transfer time should obey (1)]. The low-frequency jitter range is set from 30 to 500 Hz, which is detected by the high-frequency sensor on Yaogan-26. The high-frequency jitter range is empirically set from 600 to 2000 Hz, higher than the fundamental frequency of 504.05 Hz (the inverse of the product of the charge transfer time and TDI-CCD stage). We use an edge image [Fig. 13(a)] to analyze the degraded image quality and calculate the MTF for every ten from the 10th row to the 250th row based on ISO12233 [60] (in some images, we calculate MTF for every 20 rows to obtain an accurate result).

1) *Low-Frequency Jitter Degradation Analysis:* We degraded the image in the across-track direction and along-track direction separately using the jitter curves $D_p(t) = 1.5\sin(2\pi \times 300 \times t)$. The simulation result is shown in Fig. 14(a)–(d). Low-frequency jitter will not only blur the image but also cause geometric distortion, and across-track direction jitter creates more severe degradation. The MTF for each row is obviously row-variant, which matches the conclusion in previous research [16].

[1]Trademarked.

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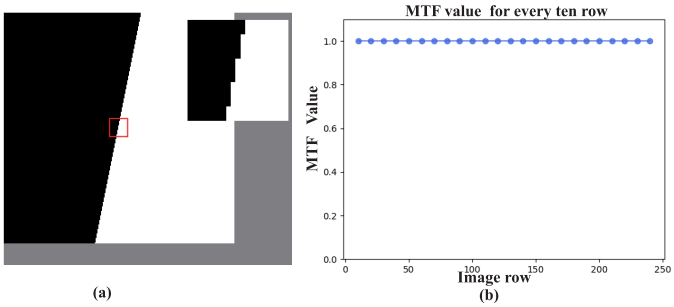


Fig. 13. (a) Testing label and partly magnified images. The image is shown with a gray background to make the white part visually obvious. (b) MTF value for every ten rows of label image at 0.2 normalized frequency. As an ideal edge, the MTF value is 1 for every ten rows.

2) *High-Frequency Jitter Degradation Analysis:* Similar to the low-frequency jitter degradation analysis, we degrade the image using $D_p(t) = 1.5\sin(2\pi \times 80 \times t)$ in two directions, as shown in Fig. 14(e)–(h). It is shown that the high-frequency jitter curves blur the image but do not cause obvious distortion. This is partly because the jitter curves change rapidly, and at every charge transfer time, the MAA of the jitter curves [see (8)] are approximately the same. The MTF curves show that the blur caused by high-frequency jitter is spatially invariant.
3) *Charge Transfer Time Mismatch Degradation Analysis:* Considering the charge transfer time mismatch degradation, we separately set the mismatch α to 2%, 10%, and 30%. As shown in Fig. 14(i)–(n), the mismatch will compress the label image. The blur is also spatially variant.
4) *Drift Angle Simulation:* We set the drift angle to 2°and 10° Fig. 14(o)–(r) shows the result. We can see obvious geometric distortion. The blur varies slightly for each row.
5) *Coupled Degradation:* In most real imaging situations, the jitter curves are not ideally sinusoidal functions, and all these degradation factors couple together, making obviously row-variant blurred images. Fig. 14(s) and 14(t) shows the images degraded by the coupled factors.
6) *The TDI-CCD Stage Effect:* The TDI camera can obtain more energy for a single object with more TDI-CCD stages, but the degradation accumulates more due to charge accumulation. Under the same degradation parameters, the image quality varies with different TDI-CCD stages and is shown in Fig. 15. When the TDI-CCD stage is equal to 1, the TDI camera becomes the original linear-push-broom camera, and the blur caused by low-frequency jitter is not obvious, but the distortion becomes more severe. With the increase in the TDI-CCD stage, the distortion becomes less obvious, but the blur is more severe. In addition, the blur is closer to spatially invariant when the TDI-CCD stage is larger, which can be explained by the fact that with the increase in the TDI-CCD stage, the variance in the MAA [see (8)] of each row is lower.

B. Dataset Preparation

The learning-based restoration method depends heavily on the training dataset. In this experiment, we choose PatternNet [61] as the label image. PatternNet is a large-scale high-resolution remote sensing dataset collected from Google Earth imagery or the Google Maps API. It has 38 classes, and each class has 800 images, sufficient for training.

We first convert all the images into gray images to represent the original image in one optical band. The restoration of multispectral degraded images will be discussed in future work; the reason is explained in the Supplementary Material. We generate the degraded images using the following parameters, which partly refer to the parameters of the Yaogan-26 satellite and very high-resolution satellite telescopes (VHRSTs) [8], [43].

1) The simulated image has the same size as the label image.
2) The TDI-CCD stage is set to 32, the photogenic charge transfer time is set to 61.998 μs, and the time sampling rate inside one photogenic charge transfer time is set to 40, which exceeds the Nyquist frequency of the high jitter frequency.
3) The x-direction and the y-direction jitter curves are composed of four sinusoidal functions. The amplitudes are randomly selected in [0.3, 2]. Two sinusoidal frequencies are randomly selected in [30, 500] Hz, and the others are randomly selected in [500, 2000] Hz. The phase is randomly selected in [0, 2π].
4) Photogenic charge transfer time randomly deviates from normal by [−2%, 2%], and the deflection angle is randomly set to [−2°, 2°].

It should be emphasized that the training dataset and test dataset similarity directly affect the restoration methods. If the degradation mechanism of the training dataset deviates from the mechanism of the test dataset, the restoration methods will not behave well, as shown in Fig. 16. Obviously, the networks can partly deblur the images but cannot correct the geometric distortion. Therefore, the simulation method for the TDI camera, which produces a sufficient paired dataset, is essential for learning-based degraded image restoration.

For each class, the ratio of the training dataset, validation dataset, and test dataset is set to 6:2:2.

C. Restoration Results on Simulated Datasets

We use PyTorch to implement our restoration network and train it on a single NVIDIA RTX A6000 for 300 epochs. In network training, the image is fed to the network at full size (256 × 256), and the batch size is set to 4 for each iteration. We use the Adam optimizer with an initial learning rate of 2×10^{-4}, which is steadily decreased to 10^{-6} using the cosine annealing strategy [63]. We compare our methods with some representative iterative-optimized restoration methods, including unconstrained neural optimization solution to blind deconvolution (Self-Deblur) [22], image restoration with mean-reverting stochastic differential equations (IR-SDE) [30], learning deep CNN denoiser prior for

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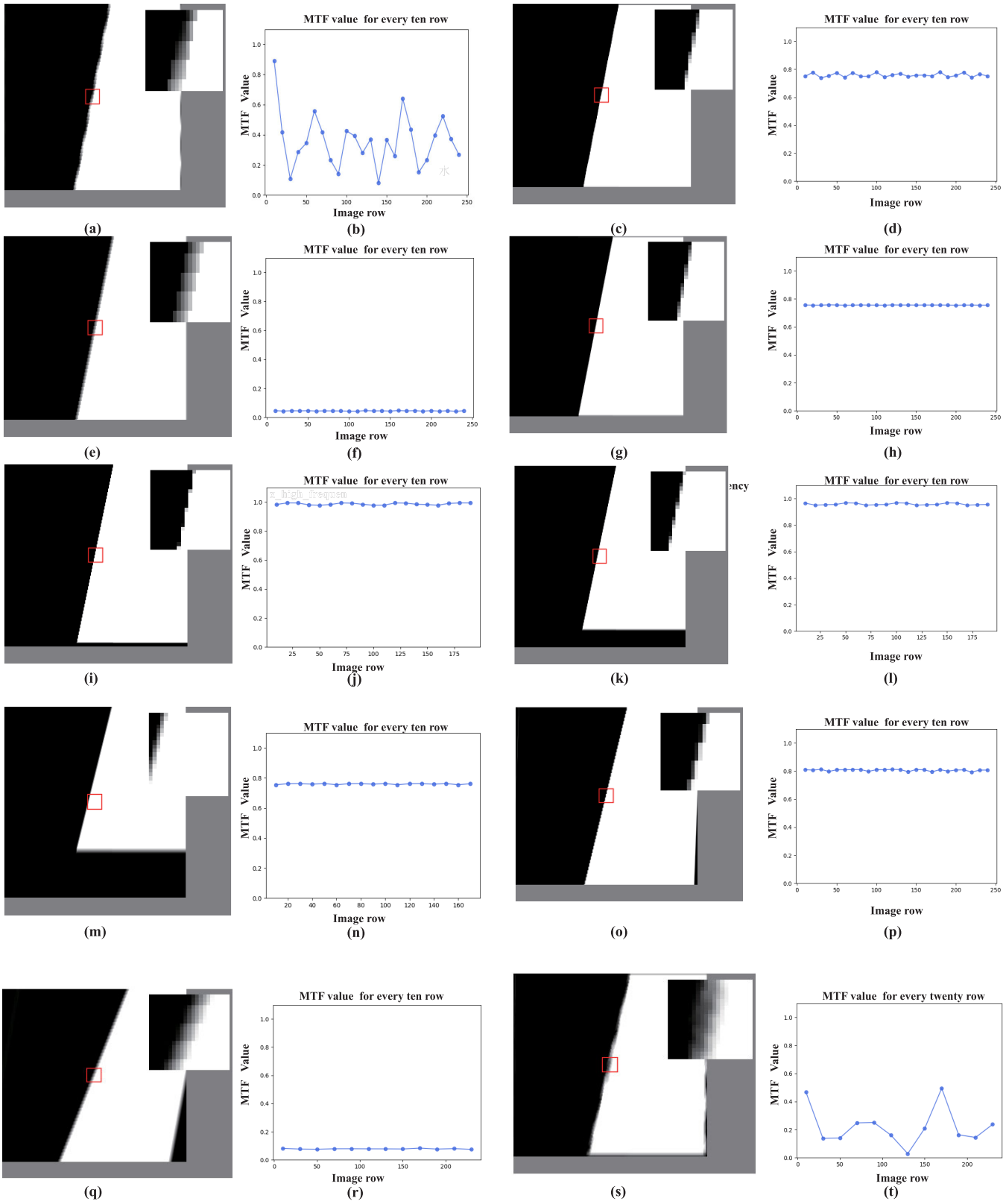


Fig. 14. (a) Edge image degraded by jitter curve $D_{px} = 2\sin(2\pi \times 300 \times t)$ and (b) MTF curves. (c) Edge image degraded by jitter curve $D_{py} = 2\sin(2\pi \times 300 \times t)$ and (d) MTF curves. (e) Edge image degraded by jitter curve $D_{px} = 2\sin(2\pi \times 2000 \times t)$ and (f) MTF curves. (g) Edge image degraded by jitter curve $D_{py} = 2\sin(2\pi \times 2000 \times t)$ and (h) MTF curves. (i) Edge image with a charge transfer time mismatch of 2% and (j) MTF curves. (k) Edge image with a charge transfer time mismatch of 10% and (l) MTF curves. (m) Edge image with a charge transfer time mismatch of 30% and (n) MTF curves. See the Supplementary Material for the image with charge transfer time mismatch -2%, -10%, and -30%. (o) Edge image with a drift angle of 2° and (p) MTF curves. (q) Edge image with a drift angle of 10° and (r) MTF curves. See the Supplementary Material for the drift angle of $-2°$ and $-10°$. (s) Edge image degraded by coupled factors including jitter $D_{px}(t) = 2\sin(2\pi \times 300 \times t) + 1.2\sin(2\pi \times 400 \times t + 1) + \sin(2\pi \times 1000 \times t + 2) + 0.8\sin(2\pi \times 1200 \times t + 3)$, $D_{py}(t) = 2\sin(2\pi \times 300 \times t + 4) + 1.2sin(2\pi \times 415 \times t + 5) + \sin(2\pi \times 1000 \times t + 7) + 0.8\sin(2\pi \times 1200 \times t + 8)$ curves, charge transfer time mismatch of 1%, drift angle of 1°, and (t) MTF curves.

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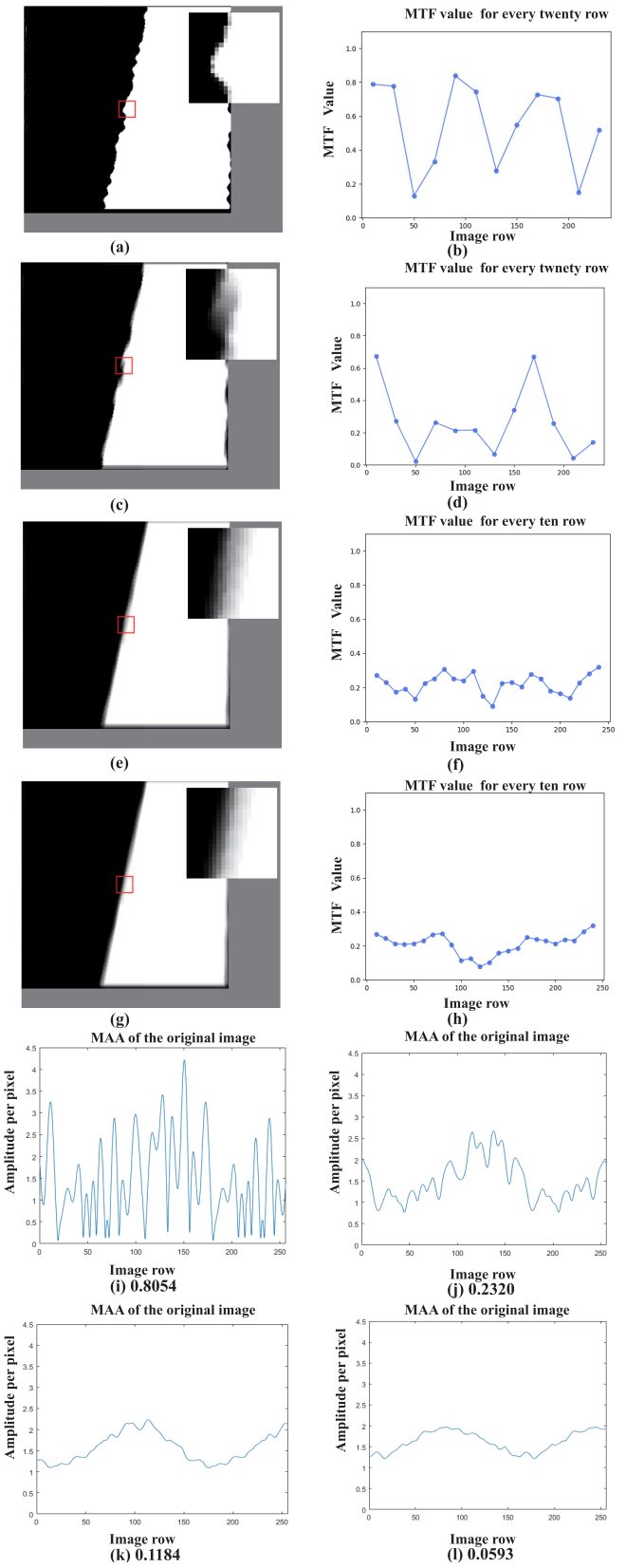


Fig. 15. Edge image degraded by coupled factors with the same parameters as in Fig. 14(s) with different TDI-CCD stages. (a) TDI-CCD stage is set to 1 and (b) MTF curves. (c) TDI-CCD stage is set to 16 and (d) MTF curves. (e) TDI-CCD stage is set to 16 and (f) MTF curves. (g) TDI-CCD stage is set to 64 and (h) MTF curves. (i) TDI-CCD stage is set to 96 and (j) MTF curves. (k)–(l) Calculated MAAs for each row. With the increase in the TDI-CCD stage, the variance in the MAA becomes lower, and the row MTF is smoother.

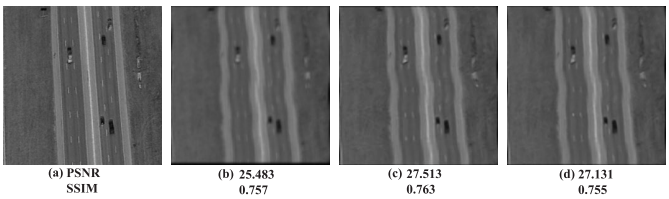


Fig. 16. Result of the restoration method trained on the GoPro dataset [53] and tested on our simulation degraded TDI dataset, noting that the GoPro dataset assumes motion blur, which is different from TDI degradation. (a) Label image. (b) Simulated image. (c) Restoration result of MIMO-UNet [25]. (d) Restoration result of MPR-Net [62]. The metrics below the image are the PSNR and SSIM. The metrics in the following figures are the same.

TABLE I
QUANTITATIVE RESULTS OF DIFFERENT METHODS

Method	Average PSNR	Average SSIM	FLOPs
Degraded	23.605(57.1%)	0.658(65.5%)	/[1]
IRCNN	18.782(59.5%)	0.498(76.5%)	/
Self-Deblur	23.112(59.3%)	0.630(68.1%)	/
DPDNN	27.602(32.1%)	0.801(40.7%)	298.240G
Deblur-Gan-v2	24.728(51.2%)	0.717(58.3%)	83.089G[2]
XY-Deblur	28.620(23.7%)	0.839(26.7%)	150.26G
MIMO-Unet	29.072(19.6%)	0.839(26.7%)	397.854G[3]
IR-SDE	25.827(44.7%)	0.743(54.1%)	/
Restormer	29.827(12.3%)	0.861(15.1%)	281.981G
Our method	30.970(0.0%)	0.882(0.0%)	102.988G

[1] We do not include FLOPs of iterative-optimized methods, as they commonly require a relatively long time to restore degraded images.
[2] We use Inception-ResNet-v2 as the backbone of deblur-GAN-v2.
[3] We use MIMO-UNet-plus in the official implementation.

image restoration (IRCNN) [32], and discriminative methods, including denoising prior driven deep neural network for image restoration (DPDNN) [24], MIMO-Unet [25], Deblur-Gan-v2 [26], Restormer [27], and XY-Deblur [31]. Table I shows the statistical analysis of average peak signal-to-noise ratio (PSNR) and average structural similarity (SSIM) results for the comparison methods on the testing datasets.

Among the restoration methods, our restoration method achieves the best average PSNR and SSIM. In addition, for the model complexity, we compare the floating-point operations (FLOPs) of different discriminative models (we do not include the FLOPs of iterative-optimized methods because they take a relatively long time to iterate). The results show that our restoration model has a lower computational burden than the competitive methods instead of Deblur-gan-v2. Fig. 17 shows the restoration results of different methods; we analyze them in the following.

Self-Deblur and IRCNN are classic model-based restoration models. The Self-Deblur method aims to estimate the PSF of the whole image and then iteratively recover the image based on the estimated PSF. IRCNN needs the preestimated PSF as input (as we do not know the PSF without assistance information, we use the Gaussian blur kernel as input because it is widely adopted in other research) and uses half-quadratic splitting methods to optimize the objective function to recover the degraded images. However, these two methods ignore the fact that the degraded images have spatially variant PSFs, and the restoration results are not satisfactory. The recovered images

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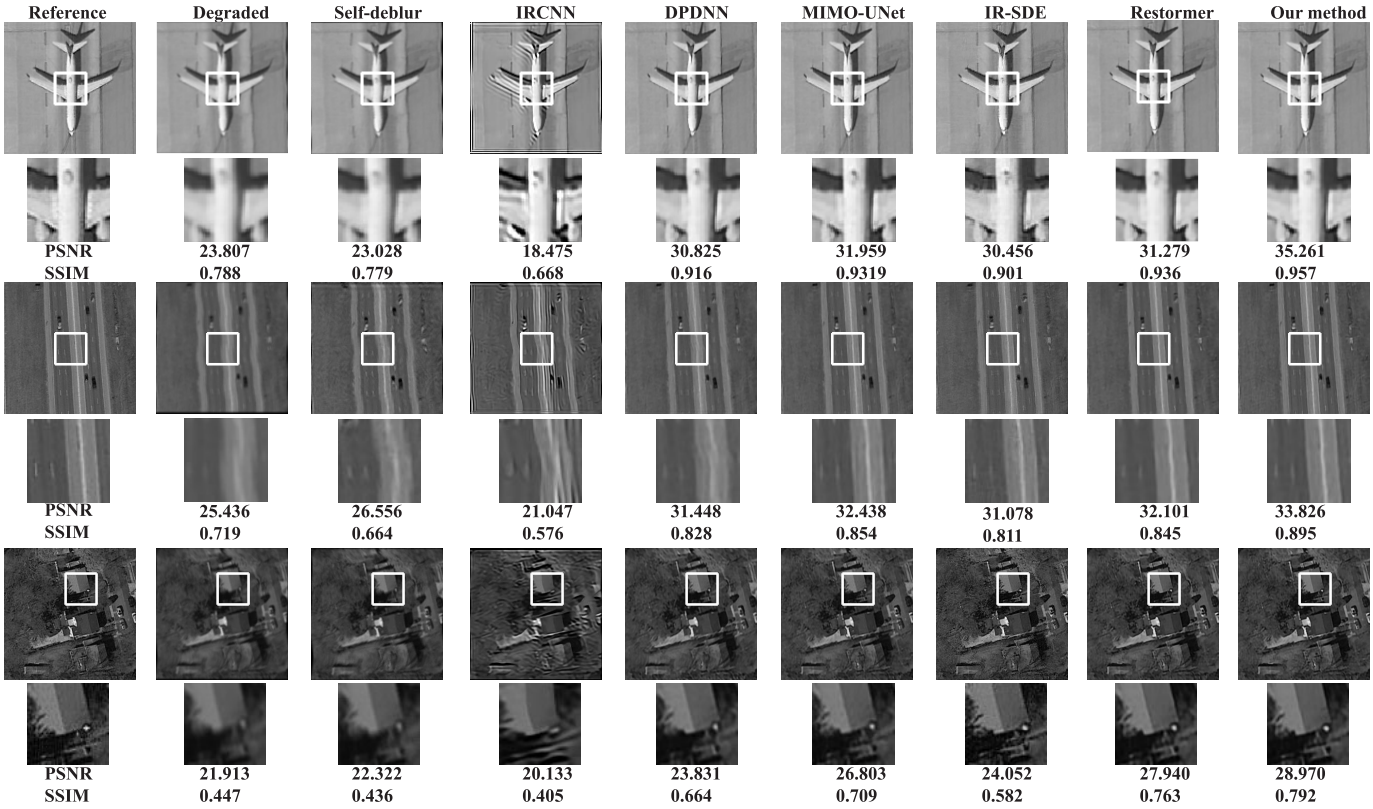


Fig. 17. Some restoration results of different methods. The partly amplified images and metrics (PSNR and SSIM) are also listed. The results of Delbur-Gan-v2 and XY-Deblur are not listed here to save space. For more comparative images, see the Supplementary Material.

are still geometrically distorted, and the images recovered by IRCNN suffer from the ringing effect. IR-SDE is a state-of-the-art restoration method that first models image degradation using stochastic differential equations (SDEs) and then recovers the degraded images by simulating the corresponding reverse-time SDE. Although IR-SDE can effectively deblur the image, the geometric distortion is still visible. We attribute this to the fact that the degradation of TDI cameras is more complicated than traditional degradation, including Gaussian noise, motion deblurring, and rain, as mentioned in the original IR-SDE.

Restormer is a special kind of transformer that uses a multideconv head to avoid quadratic computation in traditional transformers. It achieves satisfactory results in image restoration, although with slight color distortion. Its computational burden is higher than that of convolution networks. DPDNN, MIMO-UNet, and our restoration method are all discriminative methods that adopt the classic UNet architecture as the backbone but with different blocks. These methods all deblur the image and correct the geometric distortion. The results show that compared to DPDNN and MIMO-UNet, our network can reproduce detailed textures and achieve better visual performance.

D. Restoration Results of Yaogan-26 Remote Sensing Images

To further verify the effectiveness of our proposed method, we select some optical remote sensing images captured by the Yaogan-26 remote sensing satellite. These images are captured by a TDI camera with an integration stage of 24, and an exposure time is 61.998 μs. In addition, this satellite is equipped with a high-frequency angular accelerator, and it has been reported to suffer from jitter with frequencies of 30 and 100 Hz. The captured images suffer from obvious geometric distortion due to jitter. Although the geometric distortion can be compensated for using the detected jitter curves acquired by an angular accelerator [15], not all satellites carry such precision equipment, and in some cases, we have to use blind restoration methods to restore the degraded images captured by TDI cameras.

We consider the compensated images as label images and input the real-world degraded images to different restoration methods. It should be emphasized that all the restoration methods listed below do not know the previously detected jitter curves. The restoration results are listed in Fig. 18.

Obviously, iteratively optimized methods, including IRCNN, Self-Deblur, and IR-SDE, cannot correct the geometric distortion. Similar to experiments on simulated datasets, iterative methods commonly assume spatially invariant blur, which cannot accurately reflect the degradation factors of TDI cameras, so poor restoration results are expected. For the learning-based methods, Restormer does not perform well in correcting the real-world geometric distortion. In some parts of the images, the blur is more severe than in degraded images. UNet architectures, including DPDNN, MIMO-UNet, and our method, all obviously correct the distortion. However, DPDNN and MIMO-UNet smooth the original images, and the recovered images lose some details.

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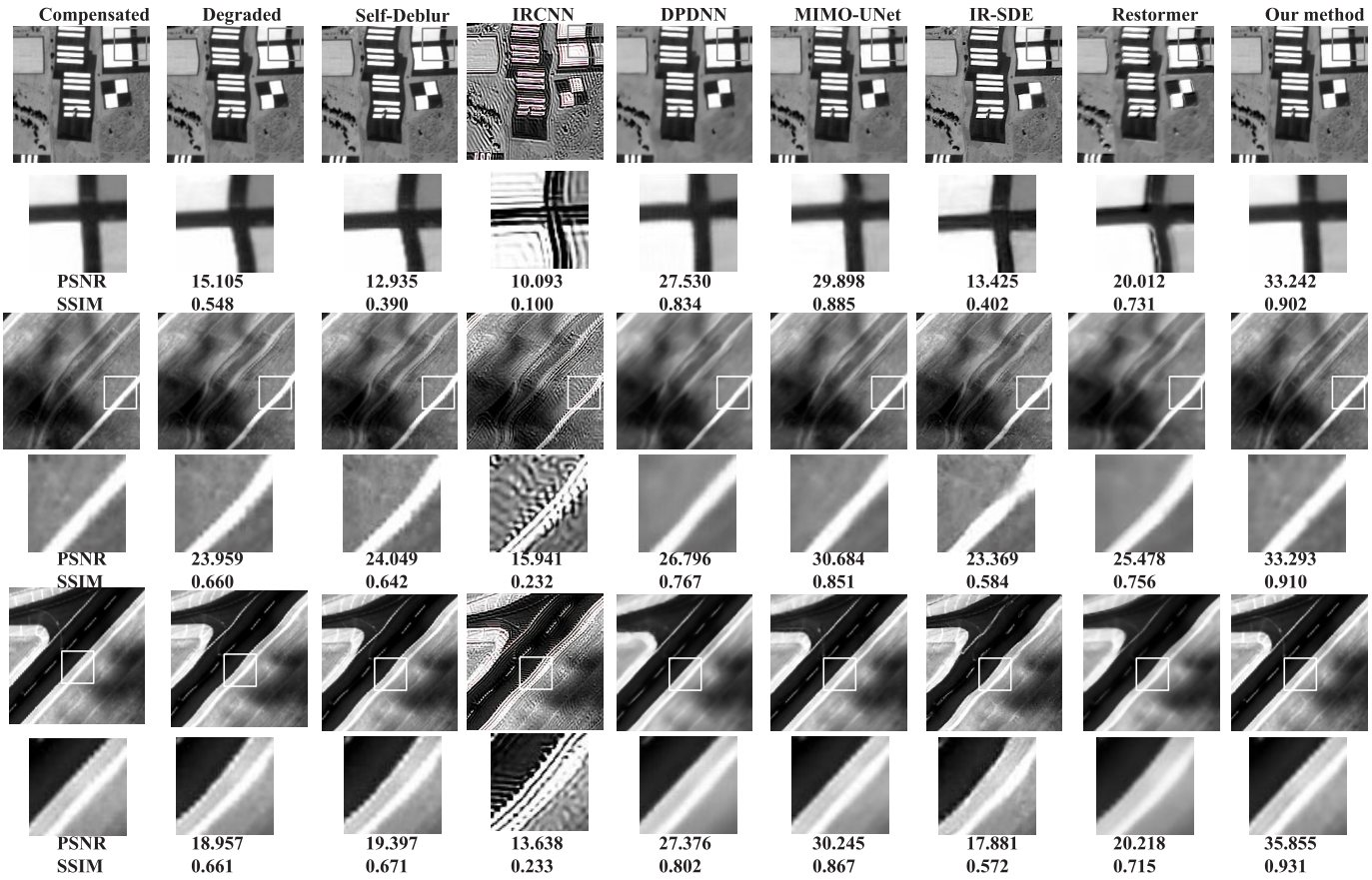


Fig. 18. Restoration results on real-world images captured by the Yaogan-26 satellite. The partly amplified images and metrics (PSNR and SSIM) are also listed.

TABLE II
ABLATION STUDY OF DIFFERENT BLOCKS

Ablation study	Average PSNR	Average SSIM	FLOPs
Row-attention block	30.583(4.3%)	0.877(4.1%)	93.316G
Row-encoder block	30.649(3.6%)	0.878(3.3%)	99.368G
Channel attention block	30.094(9.0%)	0.867(11.2%)	98.257G
Spatial attention block	30.730(2.1%)	0.879(2.5%)	98.251G
Context block	30.850(0.7%)	0.879(2.5%)	102.829G
AFF	30.741(2.0%)	0.879(2.5%)	97.083G
Change stage to 3	30.649(3.0%)	0.878(3.3%)	91.228G
Compress the basic block	30.740(2.0%)	0.879(2.5%)	84.198G
Combine row-attention block into basic block	30.381(6.0%)	0.871(8.5%)	93.315G
full network	30.970(0.0%)	0.882(0.0%)	102.988G

In contrast, our method can preserve the details of the original images while correcting the geometric distortion. It can effectively recover degraded images without detailed information about jitter curves.

E. Ablation Study and Parameter Study

To show the effectiveness of our proposed restoration network, we perform an ablation study of different modules on the simulation datasets. The statistical results on the test dataset and the FLOPs of the ablation network are shown in Table II.

1) Ablation on Row-Encoder Block and Row-Attention Block: As discussed in Section IV, the degraded images captured by TDI cameras are row-variant blur, so we introduce row-encoder block and row-attention module to help the network be aware of the features. We remove these blocks and show that the row-attention block can improve the average PSNR by approximately 0.387 dB and that the row-encoder block can improve the average PSNR by approximately 0.321 dB.

2) Ablation on the Channel Attention Block: We remove the channel attention module in the basic block and show that the average PSNR drops highly in the test datasets. Adding the channel attention block will not bring much burden to FLOPs, but it can obviously improve the restoration ability.

3) Ablation on the Spatial Attention Module: Similar to the ablation study on the channel attention module, the spatial attention module in the basic block is removed to carry out the ablation study. The spatial attention module will help the

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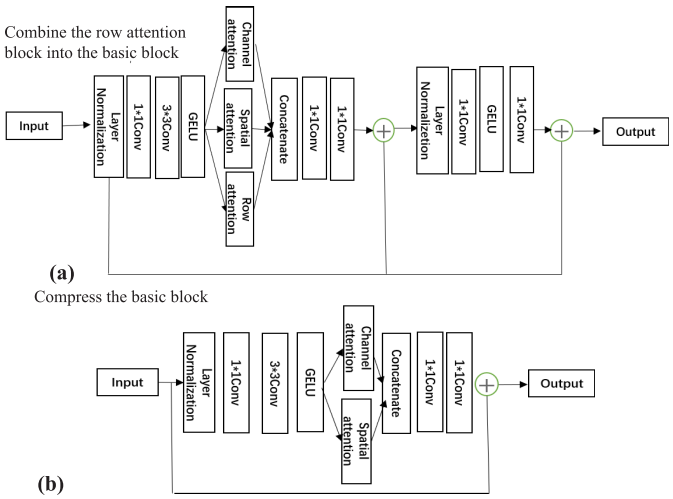


Fig. 19. (a) We attempt to combine the row-attention block into the basic block. (b) We attempt to remove the latter part of the basic block.

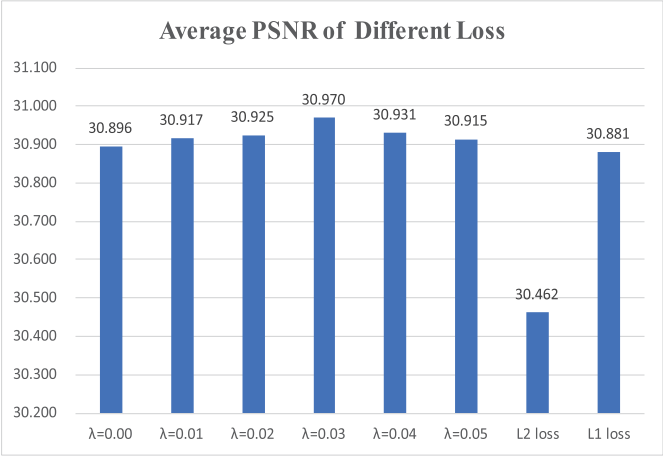


Fig. 20. Analysis of different loss functions.

network identify the spatially variant blur and improve the average PSNR by approximately 0.240 dB.

4) Ablation on AFF: The AFF block fuses features from different stages and helps information flow from multiple directions instead of flowing only from bottom to top. We remove the AFF blocks and deliver only the feature maps between adjacent stages, showing that AFF blocks can help increase the average PSNR by approximately 0.174 dB. Although it seems that many dense connections are connected to the AFF block, the relative FLOP increase is approximately 6%, and we think that the improvement is worthwhile compared to the restoration ability improvement.

5) Ablation on Context Block: The context block is placed between the lowest encoder and decoder, which helps the network improve its receptive field. We change the context block to a traditional convolution block to conduct an ablation study, showing that the context block can improve the average PSNR by approximately 0.12 dB. Although the increase is relatively small compared with other blocks, the relative FLOP increase is very small (approximately 0.15%).

6) Choice of the Network Architecture: We also attempt to use other network architectures to validate the effectiveness of our network, including decreasing the encoder and decoder stages from 4 to 3, combining the row-attention block into the basic block, and compressing the network's basic block. The new basic block is shown in Fig. 19. The restoration result shows that the proposed network outperforms other network architectures.

The most important hyperparameter in our proposed network is the loss hyperparameter λ, which balances the Charbonnier and edge losses. We fine-tune the hyperparameter in experiments and find that setting λ to 0.03 obtains the best results. We also test other losses, including the traditional $L2$ loss and $L1$ loss, and the results show that the performance is poor compared to the fine-tuned loss. The results are shown in Fig. 20.

VI. Conclusion

In this article, we proposed an image-to-image TDI camera simulation method and a learning-based method to restore degraded images affected by different factors, including jitter curves, photogenic charge transfer time mismatch, and drift angle. The simulation results correspond to other previous work. It is also a more comprehensive method than the other image-to-image simulation methods as it considers the degradation coupled factors in one simulation process instead of simulating the degradation in sequence. Moreover, the difficulty in acquiring remote sensing ground-truth images has been a significant obstacle. Our method for simulating TDI cameras' imaging mechanism helps to provide physically based image pairs and solve this obstacle. The proposed restoration method can effectively correct the distortion and deblur the degraded images, outperforming other competing restoration methods. Also, this article is set to serve as a bridge linking the field of TDI cameras for remote sensing and learning-based image restoration methods. Restoring more real-world degraded images will be the direction of our future research.

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Menghao Li received the B.S. degree from Nanjing University, Nanjing, China, in 2021. He is currently pursuing the Ph.D. degree with Zhejiang University, Hangzhou, China.

His research interests include remote sensing imaging simulation and remote sensing image restoration.



Ziran Zhang received the B.S. degree from Central South University, Changsha, China, in 2020. He is currently pursuing the Ph.D. degree with Zhejiang University, Hangzhou, China.

His research interests include image restoration and remote sensing imaging systems.



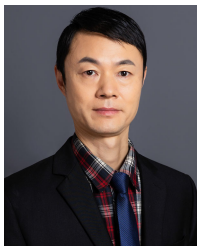
Shiqi Chen received the B.S. degree from the Huazhong University of Science and Technology, Wuhan, China, in 2018. He is currently pursuing the Ph.D. degree with the State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University, Hangzhou, China, supervised by Prof. Huajun Feng and Prof. Zhihai Xu.

His research interests include computational photography and physical-based vision.



Zhihai Xu received bachelor's, master's, and Ph.D. degrees from Zhejiang University, Hangzhou, China, in 1986, 1989, and 1996, respectively.

He is currently a Professor with the State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University. His research interests include optical remote sensing and camera imaging chains.



Qi Li received the Ph.D. degree in optical engineering from Zhejiang University, Hangzhou, China, in 2004.

He is currently a Professor with the State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University. His research interests include optical system design and imaging techniques.



Huajun Feng received the B.S. and M.S. degrees from Zhejiang University, Hangzhou, China, in 1983.

He is currently a Professor with the State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University. His research interests include imaging techniques, imaging processing, precision testing technology, and optical system design.



Yueting Chen received the bachelor's and Ph.D. degrees from Zhejiang University, Hangzhou, China, in 2004 and 2009, respectively.

He is currently a Lecturer with the State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University. His research interests include computational imaging and optical imaging.

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